

Simple Harmonic Motion

Building Models to Describe our World



Outline

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- The Concept and Formula

- Application example: spring-mass system

- Integration Constants

- SHM in relation to rotational motion

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- SHM is everywhere!

The Structure of SHM

Simple Harmonic Motion

Simple Harmonic Motion

- Anytime that you have this equation of motion:

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

- We call the motion “Simple Harmonic Motion” and the solution is:

$$x(t) = A \cos(\omega t + \phi)$$

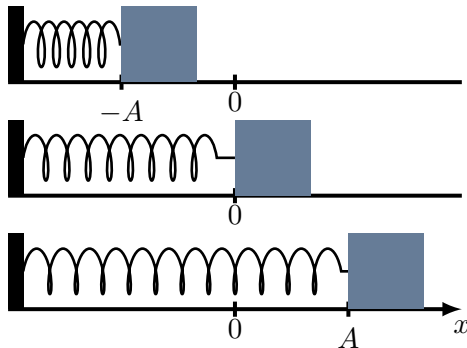
(note that sine also works)

- Where A and ϕ are “integration constants” that depend on the specific “initial conditions”.

Applications of SHM: spring-mass system

- The equation of motion for SHM comes up *a lot* in physics.
- The most obvious case is to describe the position of a mass attached to a spring:

$$\begin{aligned}ma &= -kx \\ m \frac{d^2x}{dt^2} &= -kx \\ \therefore \frac{d^2x}{dt^2} &= -\frac{k}{m}x = -\omega^2x \\ \omega &= \sqrt{\frac{k}{m}}\end{aligned}$$



A spring-mass system is a common example of simple harmonic motion.

Applications of SHM: spring-mass system

- The position as a function of time is given by:

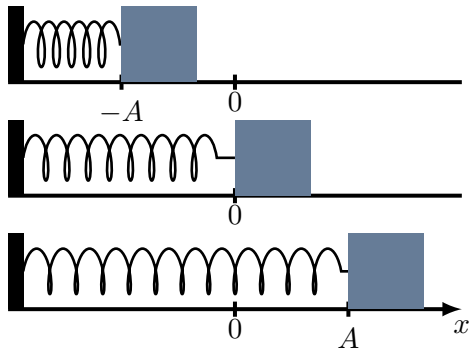
$$x(t) = A \cos(\omega t + \phi)$$

- The velocity as a function of time is given by:

$$v(t) = \frac{dx}{dt} = -A\omega \sin(\omega t + \phi)$$

- Need to determine A and ϕ from initial conditions:

→ Specify (say) position and velocity at $t = 0$.



A spring-mass system is a common example of simple harmonic motion.

Specifying the integration constants

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

- This is a second order differential equation, so we must specify 2 integration constants.
- At some time, $t = t_0$, we specify the initial position, x_0 , and initial velocity, v_0 .
- This is enough information to determine A and ϕ (say $t_0 = 0$):

$$x_0 = x(0) = A \cos(\omega t_0 + \phi) = A \cos(\phi)$$

$$v_0 = v(0) = -A\omega \sin(\omega t_0 + \phi) = -A\omega \sin(\phi)$$

- Similar to:

$$\frac{d^2x}{dt^2} = a \quad x(t) = x_0 + v_0 t + \frac{1}{2} a t^2$$

where we specify initial position, x_0 , and initial velocity, v_0

Back to rotational motion

- The position of the simple harmonic oscillator is described by the **same function** as the x or y position of a particle moving around a circle of radius A at constant angular velocity.
- The phase, ϕ , tells you where on the circle it started at time $t = 0$ (it's like θ_0).
- Depending on the context, we call, ω , the **angular frequency** or the **angular velocity** – we use the terms interchangeably.

$$x(t) = A \cos(\omega t + \phi) = A \cos(\theta(t))$$

$$\theta(t) = \phi + \omega t$$

The linear displacement of a spring in SHM is the same as x position of an object in uniform circular motion.

Simple Harmonic Motion

- Any physics model that gives:

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

is said to correspond to simple harmonic motion and is described by the same math.

- SHM arises from physics where there is a **restoring force** that is **linearly proportional to displacement**. For example, $F = -kx$.
- All of the physics is contained in ω , the angular frequency of the motion.
- For a mass on a spring:

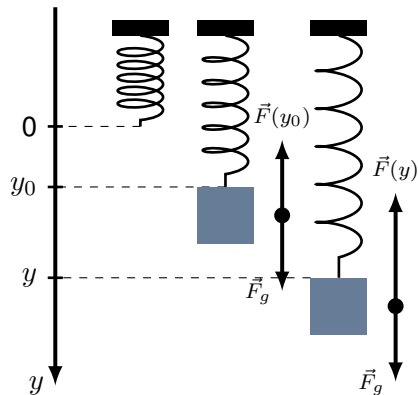
$$\omega = \sqrt{\frac{k}{m}}$$

→ The frequency of the oscillations depends on how stiff the spring is and how heavy the mass is – nothing else!

Spring Systems

Simple Harmonic Motion

Vertical mass on a spring



A vertical spring-mass system. In the middle position, the force of gravity has the same magnitude as the spring force and the system is in equilibrium.

Vertical mass on a spring

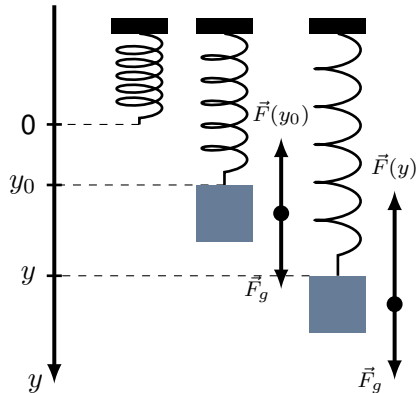
- Observations:
 - The mass oscillates about the equilibrium position, not the rest position of the spring.
 - It feels like SHM, so maybe it is?
- The math doesn't quite look like SHM:

$$mg - ky = ma_y$$

$$\frac{d^2y}{dt^2} = g - \frac{k}{m}y$$

not the same as:

$$\frac{d^2y}{dt^2} = -\frac{k}{m}y$$



A vertical spring-mass system. In the middle position, the force of gravity has the same magnitude as the spring force and the system is in equilibrium.

Vertical mass on a spring

- Consider changing coordinate system, so that the origin is located at the point about which the mass oscillates:

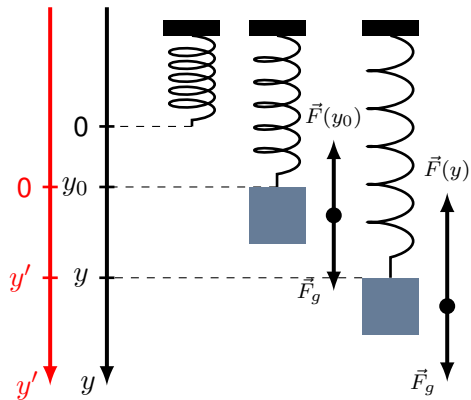
$$y' = y - y_0$$

- The extension of the spring is thus:

$$y = y' + y_0$$

- In terms of y' , Newton's Second Law reads:

$$\begin{aligned}\frac{d^2 y}{dt^2} &= g - \frac{k}{m}y \\ \frac{d^2(y' + y_0)}{dt^2} &= g - \frac{k}{m}(y' + y_0)\end{aligned}$$



A vertical spring-mass system. In the middle position, the force of gravity has the same magnitude as the spring force and the system is in equilibrium.

Vertical mass on a spring

- Starting from:

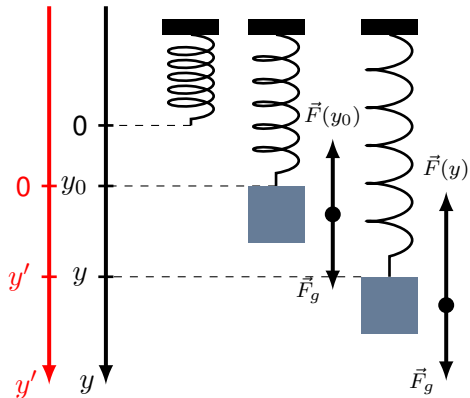
$$\frac{d^2(y' + y_0)}{dt^2} = g - \frac{k}{m}(y' + y_0)$$
$$\frac{d^2 y'}{dt^2} = g - \frac{k}{m}y' - \frac{k}{m}y_0$$

- At the equilibrium point, the spring is extended by y_0 , and the spring force is equal to gravity:

$$mg - ky_0 = 0 \quad \rightarrow \quad \frac{k}{m}y_0 = g$$

and we recover SHM using the y' axis:

$$\frac{d^2 y'}{dt^2} = -\frac{k}{m}y'$$



A vertical spring-mass system. In the middle position, the force of gravity has the same magnitude as the spring force and the system is in equilibrium.

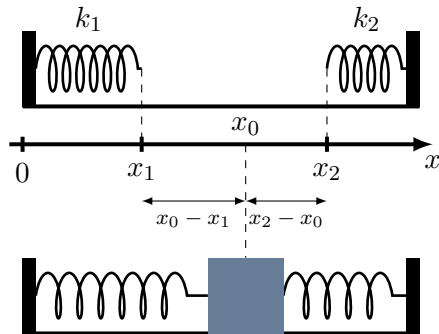
Constant force + spring force = SHM

- For a vertical spring-mass, we had an equation of the form:

$$F - kx = m \frac{d^2x}{dt^2}$$

where F is a constant force (like gravity).

- We saw that if we define a new equilibrium position (not the rest position of the spring), then we can recover simple harmonic motion about that point.
- With two springs attached to a mass, it's the same idea, and one also recovers SHM $\rightarrow$ in this case the “effective” spring constant is $k = k_1 + k_2$.



A mass connected to two springs

At equilibrium:

$$k_1(x_0 - x_1) + k_2(x_2 - x_0) = 0$$

Pendulums and Approximations

Simple Harmonic Motion

Motion of a pendulum is also SHM

- The torque from gravity (about the origin) is a *restoring torque* (it tries to bring the pendulum back to equilibrium):

$$\vec{\tau}_g = -mg \sin \theta L \hat{z}$$

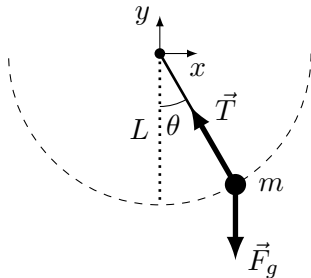
- Rotational version of Newton's Second Law:

$$\sum \vec{\tau} = \vec{\tau}_g = -mL^2 \vec{\alpha}$$

$$-mg \sin \theta L = mL^2 \frac{d^2 \theta}{dt^2}$$

$$\frac{d^2 \theta}{dt^2} = -\frac{g}{L} \sin \theta$$

not quite SHM...



Modeling a pendulum, using torque from gravity.

Motion of a pendulum is also SHM

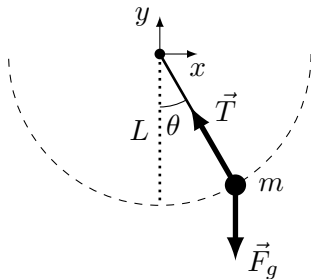
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- Rotational version of Newton's Second Law:

$$\begin{aligned}\sum \vec{\tau} &= \vec{\tau}_g = -mL^2 \vec{\alpha} \\ -mg \sin \theta L &= mL^2 \frac{d^2 \theta}{dt^2} \\ \frac{d^2 \theta}{dt^2} &= -\frac{g}{L} \sin \theta\end{aligned}$$

not quite SHM...



Modeling a pendulum, using torque from gravity.

Small angle approximation

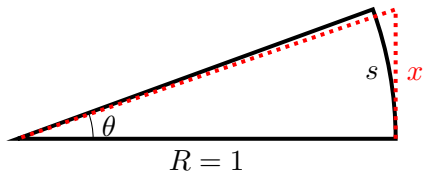
To get SHM, need to approximate $\sin \theta \sim \theta$, then:

$$\frac{d^2 \theta}{dt^2} \sim -\frac{g}{L} \theta \quad \rightarrow \quad \omega = \sqrt{\frac{g}{L}}$$

$$\therefore \theta(t) = A \cos(\omega t + \phi)$$

What does it mean for θ to be small?

- How small is small?
- You try it out on your calculator... has to be in radians for this to work...
- If the angle is too big, then it's not SHM anymore. We'll do a lab next semester to figure this out. It turns out that SHM is a very good approximation even to relatively large angles.



The small angle approximation.

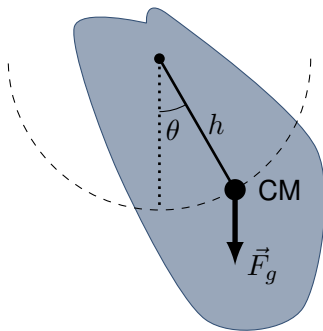
$$x = \sin \theta$$
$$s = R\theta = \theta$$

The approximation is that $s \sim x$.

Physical pendulum

- Same idea as “simple” pendulum.
- Object with moment of inertia, I , instead of a point mass:

$$\begin{aligned}\vec{\tau}_g &= I\vec{\alpha} \\ -mg \sin \theta h &= I \frac{d^2\theta}{dt^2} \\ \omega &= \sqrt{\frac{mgh}{I}}\end{aligned}$$



A physical pendulum swings because of the torque from gravity, exerted at the CM.

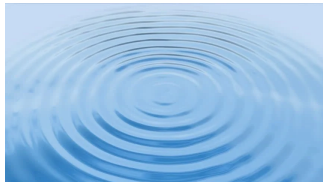
Group activity: Pendulum

What length to make the string to have a period of 1 s?

$$\omega = \sqrt{\frac{g}{L}}$$
$$T = \frac{2\pi}{\omega}$$

Simple Harmonic Motion is everywhere!

- You get SHM any time you have a “linear restoring force” (or torque).
- Everything is linear on small scale, so this can apply to many things.
- Next semester, we’ll look at waves propagating in a medium, and we’ll see that we can model the small deformations in the medium as SHM.



Surface waves on a pond can be modeled using SHM.

A wave going through a medium modeled with particles in the medium acting as simple harmonic oscillators.